

Jake Brawer, Ph.D.

[website](#) / [github](#) / [email](#) / [linkedin](#)

RESEARCH OVERVIEW

Summary I build **interactive AI systems** that combine perception, reasoning, memory, and learning to support robust human-centered interaction. My work spans **collaborative robotics** and **conversational AI**, with emphasis on realtime decision-making, structured knowledge, personalization, and deployable architectures.

PROFESSIONAL EXPERIENCE

Founding Engineer	<i>IM (voice AI / digital health)</i> Sole architect and founding engineer for an early-stage phone-based conversational AI platform for older adults. Built the end-to-end system across telephony, realtime speech, LLM orchestration, long-term memory, FastAPI services, PostgreSQL, and AWS infrastructure. Implemented platform safeguards for reliability and HIPAA-oriented handling of sensitive user data.	Nov 2025–Current
Postdoc	<i>University of Colorado, Boulder,</i> Advisors: Alessandro Roncone and Bradley Hayes Lead research on human–robot teaming, assistive robotics, and AI decision support. Develop models for inferring human proficiency, trust, and intent from gaze and behavior in collaborative environments. Build semantic robot perception systems for robust localization in cluttered indoor workspaces using low-cost visual-inertial sensing.	2023–2025
Lecturer	<i>University of Colorado, Boulder</i> Sole lecturer for Advanced Robotics (CSCI 4302/5302), a graduate-level course covering motion planning, SLAM, optimization, and reinforcement learning.	Spring 2024

EDUCATION

Yale	Ph.D. in Computer Science Advisor: Brian Scassellati	2016–2023
Vassar	B.A. in Cognitive Science; Computer Science minor	2012–2016

SKILLS

Languages	Python, C/C++, C#, SQL, Git, Emacs, Docker, Linux, $\text{\LaTeX}$
AI / ML	PyTorch, scikit-learn, SciPy, NumPy, Pandas, LangChain, LLM orchestration, RAG / vector retrieval, PGVector, matplotlib, NLTK, Gurobi, Prolog, PDDL

Backend	FastAPI, PostgreSQL, AWS, async Python, REST APIs, system design, deployment, load testing, observability, HIPAA-oriented system design, PHI-safe logging
Voice AI	Pipecat, Deepgram, AWS Bedrock, Telnix, Cartesia
Robotics/CV	ROS, Gazebo, RViz, OpenCV, Open3D, Baxter, Sawyer, UR5, Kinect, RealSense, RGB-D cameras
Soft Skills	Clear technical communication, Collaborative, Fast learner, Leadership

SELECTED PUBLICATIONS

Shivendra Agrawal, **Jake Brawer**, Ashutosh Naik, Alessandro Roncone, and Bradley Hayes (2026). “**ShelfAware: Real-Time Visual-Inertial Semantic Localization in Quasi-Static Environments with Low-Cost Sensors**”. In: *IEEE Robotics and Automation Letters*. Accepted.

Nikhil Hulle, Stéphane Aroca-Ouellette, Anthony J Ries, **Jake Brawer**, Katharina von der Wense, and Alessandro Roncone (2024). “**Eyes on the Game: Deciphering Implicit Human Signals to Infer Human Proficiency, Trust, and Intent**”. In: *33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN 2024)*. Accepted for publication, proceedings pending.

Jake Brawer, Debasmita Ghose, Kate Candon, Meiyang Qin, Alessandro Roncone, Marynel Vazquez, and Brian Scassellati (2023). “**Interactive Policy Shaping for Human-Robot Collaboration with Transparent Matrix Overlays**”. In: *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*. IEEE.

Jake Brawer, Meiyang Qin, and Brian Scassellati (2020). “**A causal approach to tool affordance learning**”. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, pp. 8394–8399.

AWARDS AND SERVICE

2025	Outstanding Reviewer Award , Robotics: Science and Systems (RSS)
2023-2025	Army Educational Outreach Program (AEOP) Fellowship , University of Colorado Boulder
2024	Area Chair, International Conference on Human–Agent Interaction (HAI)
2023	Best paper (technical track) , ACM/IEEE International Conference on Human-Robot Interaction (HRI) for “Interactive Policy Shaping for Human-Robot Collaboration”
Workshops	Co-organized the Human-Machine Teaming Paradigm Workshop (2023) and Bridging the Gap: An NSF Workshop on Conversational Agents and Human-Robot Interaction (2018).
Service	Reviewer for RSS , HRI , IROS , ICRA , Humanoids , IJRR , T-RO , and THRI ; mentored undergraduate and graduate researchers on robotics, HRI, and assistive AI projects.